8A, 20V - 100V Schottky Barrier Surface Mount Rectifier

FEATURES

- Low power loss, high efficiency
- Ideal for automated placement
- Guard ring for overvoltage protection
- High surge current capability
- Moisture sensitivity level: level 1, per J-STD-020
- RoHS Compliant
- Halogen-free according to IEC 61249-2-21

APPLICATIONS

- Switching mode power supply (SMPS)
- Adapters
- Lighting application
- Converter

MECHANICAL DATA

- Case: DO-214AB (SMC)
- Molding compound meets UL 94V-0 flammability rating
- Terminal: Matte tin plated leads, solderable per J-STD-002
- Meet JESD 201 class 2 whisker test
- Polarity: Indicated by cathode band
- Weight: 0.210g (approximately)

KEY PARAMETERS		
PARAMETER	**VALUE**	**UNIT**
I_F	8	A
V_{RRM}	20 - 100	V
I_{FSM}	150	A
$T_{J\ MAX}$	125, 150	°C
Package	DO-214AB (SMC)	
Configuration	Single die	

Pb-Free RoHS COMPLIANT HALOGEN FREE

DO-214AB (SMC)

ABSOLUTE MAXIMUM RATINGS (T_A = 25°C unless otherwise noted)								
PARAMETER	**SYMBOL**	**SK 82C**	**SK 83C**	**SK 84C**	**SK 85C**	**SK 86C**	**SK 810C**	**UNIT**
Marking code on the device		SK 82C	SK 83C	SK 84C	SK 85C	SK 86C	SK 810C	
Repetitive peak reverse voltage	V_{RRM}	20	30	40	50	60	100	V
Reverse voltage, total rms value	$V_{R(RMS)}$	14	21	28	35	42	70	V
Forward current	I_F	8						A
Surge peak forward current, 8.3ms single half sine-wave superimposed on rated load	I_{FSM}	150						A
Critical rate of rise of off-state voltage	dV/dt	10,000						V/µs
Junction temperature	T_J	- 55 to +125			- 55 to +150			°C
Storage temperature	T_{STG}	- 55 to +150						°C

THERMAL PERFORMANCE

PARAMETER	SYMBOL	TYP	UNIT
Junction-to-ambient thermal resistance	$R_{\Theta JA}$	20	°C/W

ELECTRICAL SPECIFICATIONS (T_A = 25°C unless otherwise noted)

PARAMETER		CONDITIONS	SYMBOL	TYP	MAX	UNIT
Forward voltage[1]	SK82C SK83C SK84C	I_F = 8A, T_J = 25°C	V_F	-	0.55	V
	SK85C SK86C			-	0.75	V
	SK810C			-	0.90	V
Reverse current @ rated V_R[2]		T_J = 25°C	I_R	-	0.5	mA
Reverse current @ rated V_R[2]	SK82C SK83C SK84C	T_J = 100°C	I_R	-	15	mA
	SK85C SK86C SK810C			-	10	mA

Notes:

1. Pulse test with PW = 0.3ms
2. Pulse test with PW = 30ms

ORDERING INFORMATION

ORDERING CODE[1]	PACKAGE	PACKING
SK8xC	DO-214AB (SMC)	3,000 / Tape & Reel

Notes:

1. “x” defines voltage from 20V(SK82C) to 100V(SK810C)

CHARACTERISTICS CURVES

(T_A = 25°C unless otherwise noted)

Fig.1 Forward Current Derating Curve

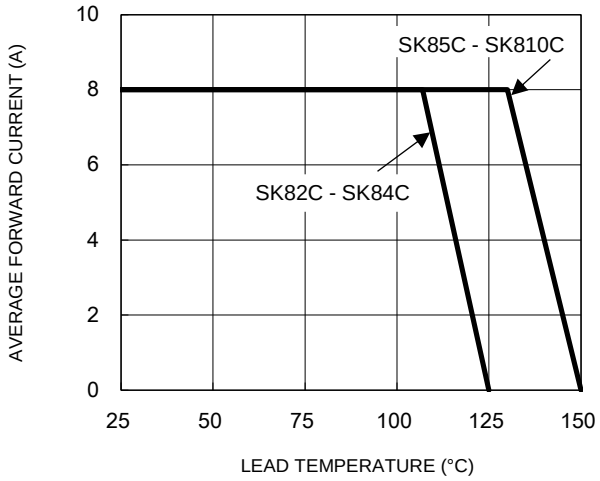


Fig.2 Typical Junction Capacitance

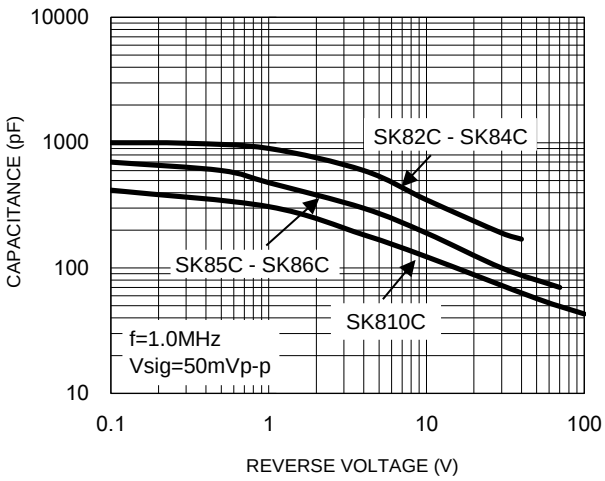


Fig.3 Typical Reverse Characteristics

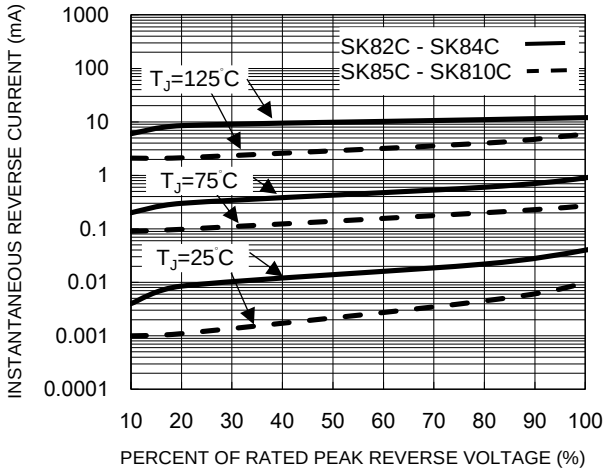


Fig.4 Typical Forward Characteristics

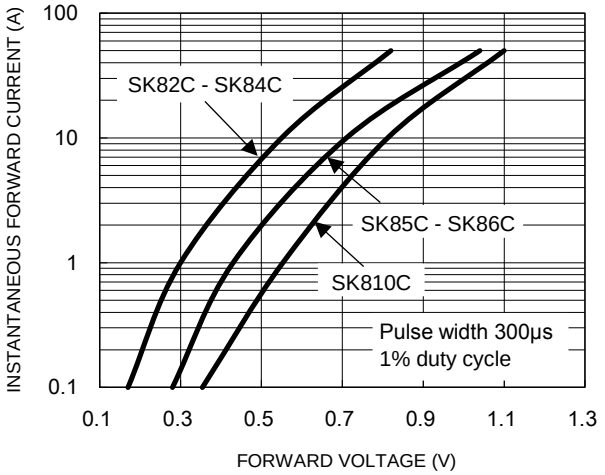
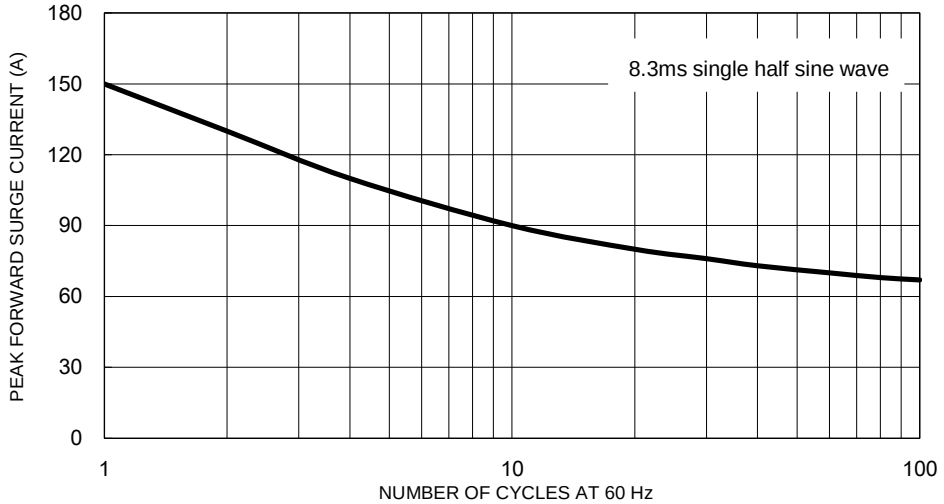


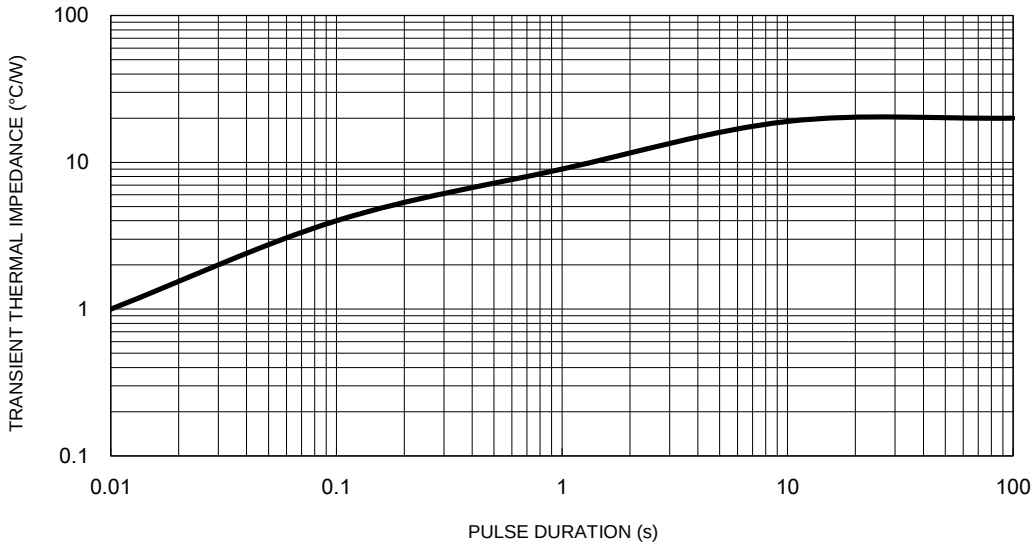
Fig.5 Maximum Non-Repetitive Forward Surge Current



CHARACTERISTICS CURVES

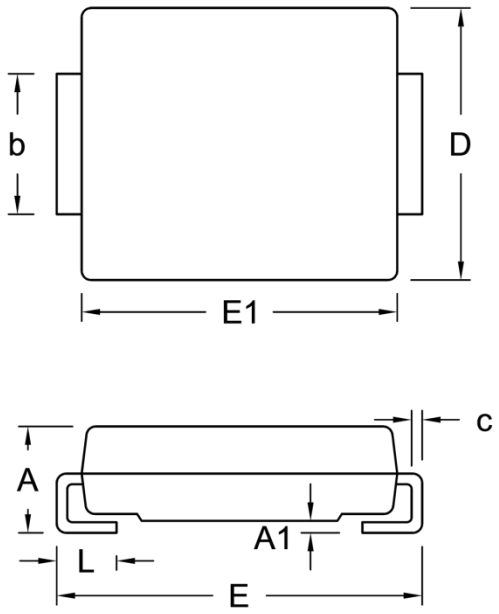
(T_A = 25°C unless otherwise noted)

Fig.6 Typical Transient Thermal Characteristics



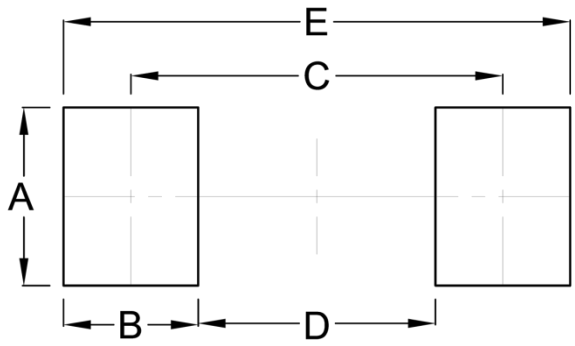
PACKAGE OUTLINE DIMENSIONS

DO-214AB (SMC)



DIM.	Unit (mm)		Unit (inch)	
	Min.	Max.	Min.	Max.
A	2.00	2.62	0.079	0.103
A1	0.10	0.20	0.004	0.008
b	2.90	3.20	0.114	0.126
c	0.15	0.31	0.006	0.012
D	5.59	6.22	0.220	0.245
E	7.75	8.13	0.305	0.320
E1	6.60	7.11	0.260	0.280
L	1.00	1.60	0.039	0.063

SUGGESTED PAD LAYOUT



Symbol	Unit (mm)	Unit (inch)
A	3.30	0.130
B	2.50	0.098
C	6.90	0.272
D	4.40	0.173
E	9.40	0.370

MARKING DIAGRAM

P/N = Marking Code
G = Green Compound
YW = Date Code
F = Factory Code

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